C.S.E. Sem -III

Session 2025-26

C.S.E 3rd Sem Live Classes New Schedule on YouTube

D.B.M.S. By Ujjwal Sir	7:00 - 8:00 AM	Monday - Saturday
Operating System By Deepak Sir	5:00 - 6:00 PM	Monday - Saturday
Others	Starting soon	Starting soon

Subjects

- Operating Systems
- Web Technologies
- Computer Network
- Computer Programming Using C
- Data base Management System

Data base Management System

• CHAPTERS/UNIT

1. Introduction(परिचय)
2. Data Model and Keys(डेटा मॉडल और कुंजियाँ)

3 Relational Model (संबंधपरक मॉडल)

4 Relational Database Design (संबंधपरक डेटाबेस डिज़ाइन)

5 SQL/MySQL

UNIT : 2 Data Model and Keys (डेटा मॉडल और कुंजियाँ)

Unit

UNIT2: Data Model and Keys: **(8 Periods)**

Define data model, Data Models : Network Model Hierarchical Model, E-R Model, Advantage & Disadvantages of each Data Model, Concept of Keys (Primary, Candidate, Super, Foreign), Constraints, Strong Entity Set and Weak Entity Set.

TOPICS

Unit ②

1. Define data model,
2. Data Models : Network Model
3. Hierarchical Model,
4. E-R Model,
5. Advantage
6. Disadvantages of each Data Model,
7. Concept of Keys (Primary, Candidate, Super, Foreign),
8. Constraints,
9. Strong Entity Set and Weak Entity Set

Data Model (डेटा मॉडल)

- A Data Model is a conceptual framework that defines how data will be stored, organized, and accessed in a database.

डेटा मॉडल एक सैद्धांतिक ढांचा (conceptual framework) है जो यह बताता है कि डेटा को कैसे संग्रहीत (store), व्यवस्थित (organize) और प्राप्त (access) किया जाएगा।

- It describes the structure, relationships, and constraints of the data.

यह डेटा की संरचना (structure), संबंध (relationships) और सीमाओं/नियमों (constraints) का वर्णन करता है।

Name
Ram
Ram

data

Students

id	Name	Branch
1	Ram	CE

Purpose of Data Model (उद्देश्य)

1. To provide a clear plan for designing the database.

   डेटाबेस डिजाइन करने के लिए एक स्पष्ट योजना प्रदान करना।

2. To help in visualizing the database structure.

   डेटाबेस संरचना को दृश्यमान बनाने में सहायता करना।

3. To define rules for storing and managing data.

   डेटा भंडारण और प्रबंधन के लिए नियम परिभाषित करना।

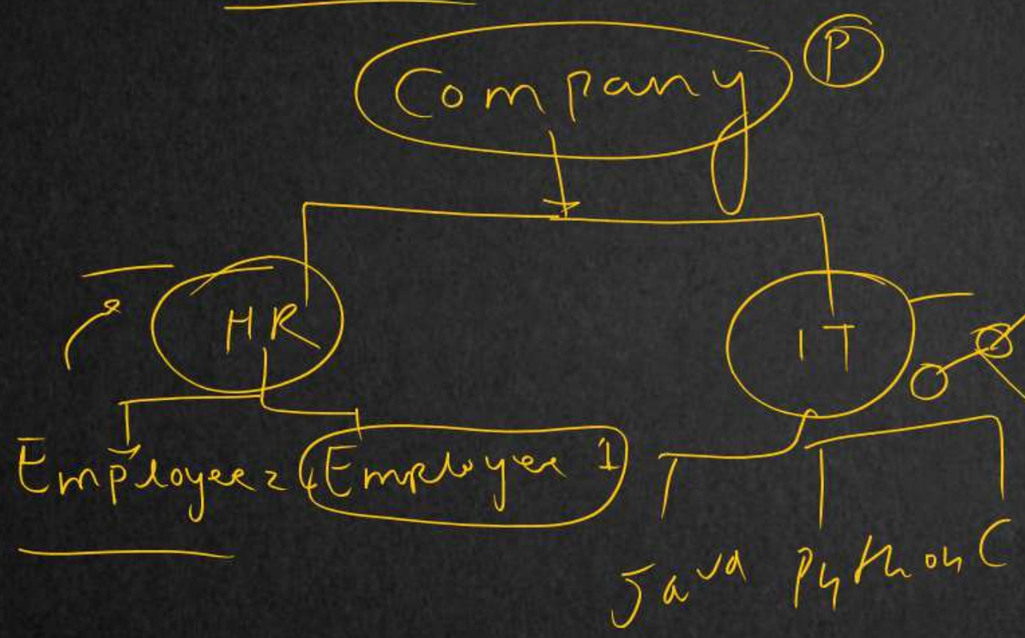
4. To maintain data consistency and integrity.

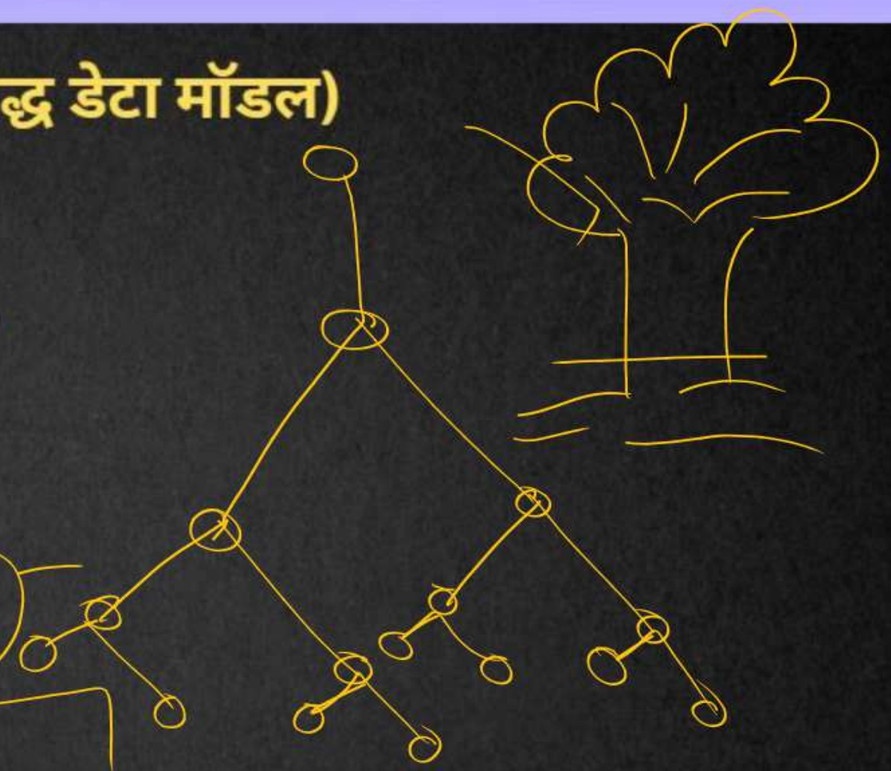
   डेटा की एकरूपता और अखंडता बनाए रखना।

Types of Data Models (डेटा मॉडल के प्रकार)

1. Hierarchical Data Model (श्रृंखलाबद्ध डेटा मॉडल)

Tree (Save)

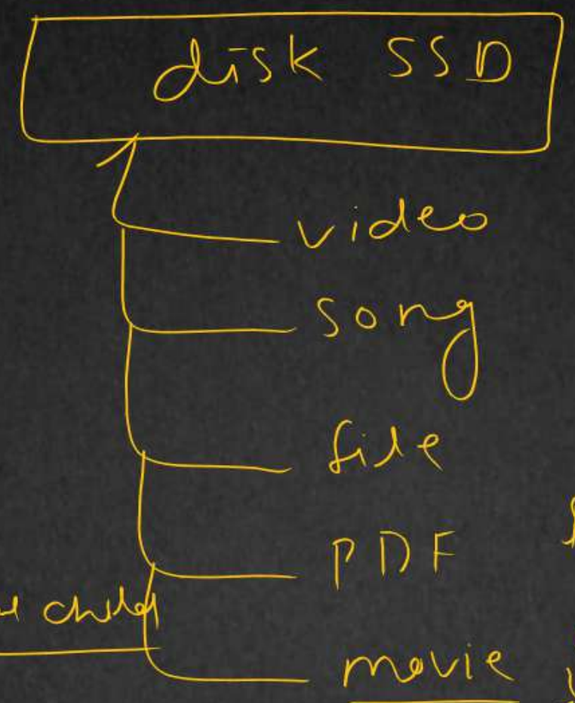


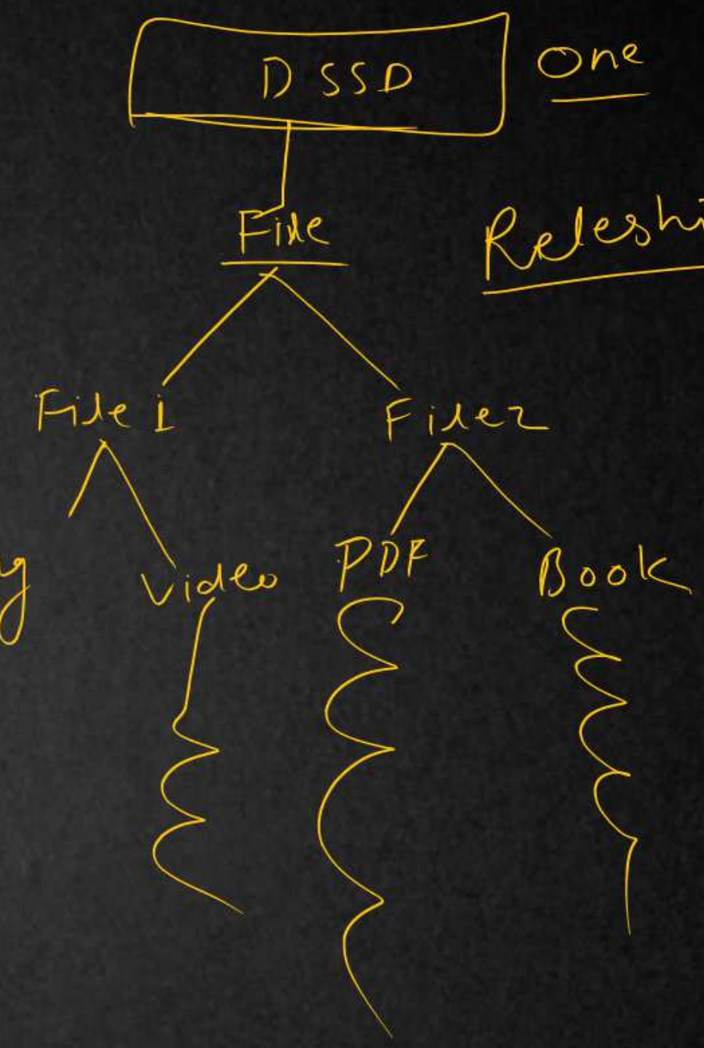


Maping

One to many

Parent to multiple child





Computer

Prent

disk e_1 D G F

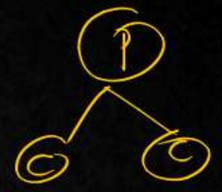
Child

File Song

- song 1
- Song 2
- Song 3
- Song 4

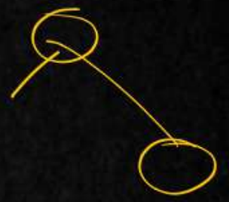
File Video

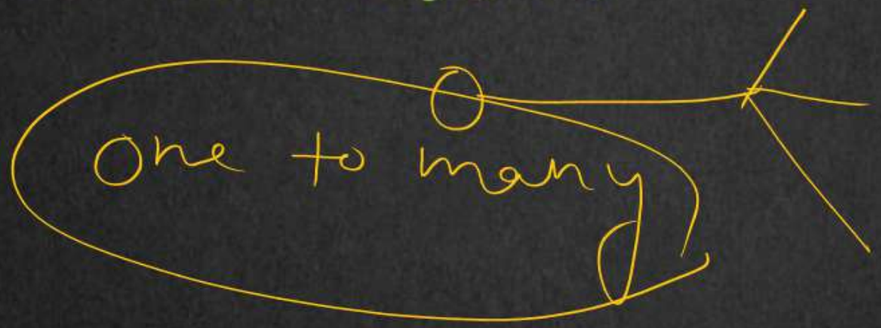
1. Hierarchical Data Model (श्रृंखलाबद्ध डेटा मॉडल)

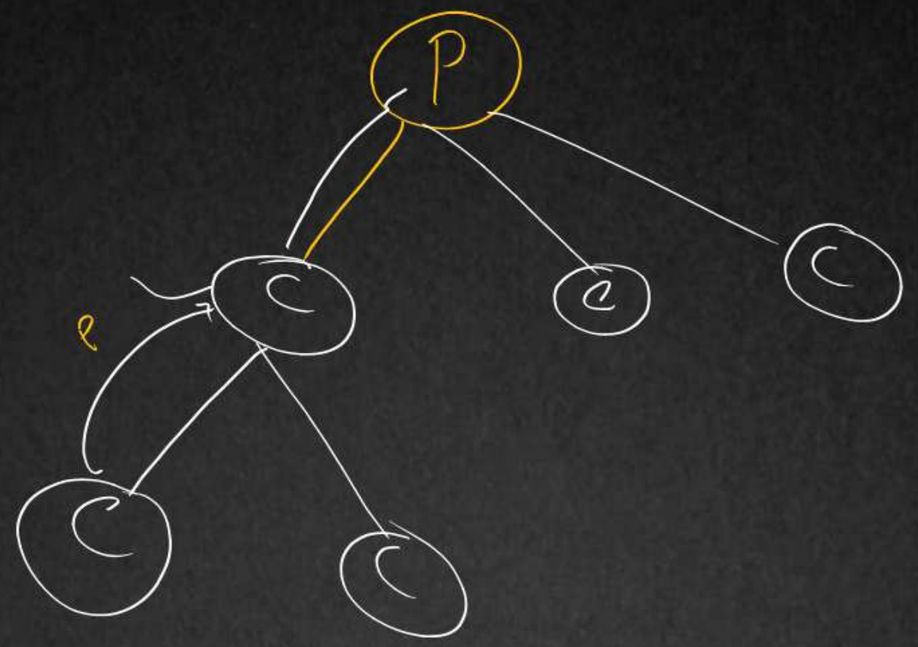


- Data is organized in a tree structure (parent-child).
  डेटा को पेड़ जैसी संरचना में रखा जाता है।
- Each parent can have multiple children, but a child has only one parent.
  एक parent के कई child हो सकते हैं, लेकिन एक child का सिर्फ एक parent होता है।
- Relationships are one-to-many.
  रिलेशनशिप one-to-many होती है।

Ex book



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Advantages:

1. Easy to understand.

   समझना आसान।

2. Good for representing data with clear hierarchy (e.g., organization chart)

   हाइरार्की वाले डेटा के लिए अच्छा।

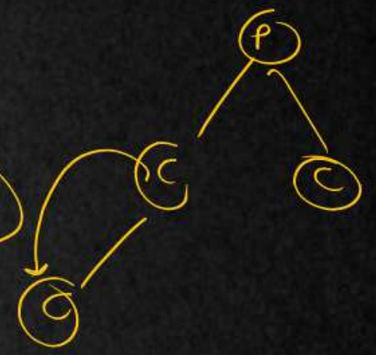
Disadvantages:

1. Cannot easily handle many-to-many relationships.

   many-to-many रिलेशनशिप कठिन।

2. Changes in structure are difficult.

   संरचना बदलना मुश्किल।



2. Network Model (नेटवर्क मॉडल)

GTech Poly Application